R16

Code No: 138GW

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, September - 2022
RENEWABLE ENERGY SOURCES
(Common to CE, EEE, ME, ECE, CSE)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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JNTUH Used Papers

1.a) Enumerate the differences between conventional and non-conventional energy sources and give examples for each.
b) Analyze the importance of renewable energy in-terms of CO_2 reduction.
c) Write a short note on types of renewable energy systems. [5+5+5]

2.a) In your view, which renewable energy source is having high potential in the Indian context? Justify your answer.
b) What is Global climate change? How is it affected because of energy sources? and list out some key points to reduce it. [7+8]

3.a) How can solar radiation be measured? With a neat diagram explain.
b) Illustrate the working principle of solar thermal energy storage systems. [7+8]

4.a) Discuss about the operation of solar Photovoltaic systems with their advantages and disadvantages.
b) Select one application of the solar energy system which you think is so important or a potential candidate. [8+7]

5.a) How does wind is classified and what are the factors influencing the wind.
b) List the types of windmills with their merits and demerits.
c) Discuss about the wind energy installations in India. [5+5+5]

6.a) Explain the advantages and disadvantages of offshore wind energy forms.
b) What are the safety and environmental aspects of wind mills?
c) List the components of a windmill and write a note on it. [5+5+5]

7.a) What are biomasses? How does a biomass conversion unit work? What are the components of a biomass conversion unit?
b) How biodiesel and alcohol are produced from biomass? [8+7]

8.a) With a neat diagram, explain the working of an OTEC plant.
b) Discuss the advantages of the wave energy conversion system. [8+7]

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